Clarifai Compute Orchestration Analysis

Customer: Hirtz and Carter **Date:** January 15, 2026

Executive Summary

This analysis provides GPU reservation recommendations for Hirtz and Carter. Clarifai Compute Orchestration handles all infrastructure - you simply reserve the capacity you need.

Full Documentation: https://docs.clarifai.com/

Your Requirements

{ "gpu_type": "", "gpu_count": "", "commitment": "" }

Project Goals: Create Image analysis system that can detect image problem (image QA)

Trending Open-Source Models (From Hugging Face)

Based on your project goals, here are recommended open-source models available on Clarifai:

Recommended Primary Models (Based on Your Use Case)

Model	Parameters	License	GPU Requirement	Link
	8B	clarifai	Standard (24GB)	mm-poly-8b ⭐

Model	Parameters	License	GPU Requirement	Link
mm-poly-8b *(Clarifai)*				
llava-1.5-7b-hf	7B	llama2	Standard (24GB)	llava-1.5-7b-hf
Llama-3.2-11B-Vision-Instruct	11B	llama3.2	Standard (24GB) or Performance (48GB)	Llama-3.2-11B-Vision-Instruct

⭐ *= Recommended Clarifai model, ready to use. Check all models at clarifai.com/explore*

Embedding Models (For RAG/Search)

Model	Parameters	License	GPU Requirement	Link
all-MiniLM-L6-v2	22M	apache-2.0	Entry (16GB) or CPU	all-MiniLM-L6-v2
bge-large-en-v1.5	335M	mit	Entry (16GB)	bge-large-en-v1.5

⭐ *= Recommended Clarifai model, ready to use. Check all models at clarifai.com/explore*

Alternative Model Options

Model	Parameters	License	GPU Requirement	Link
llava-v1.6-mistral-7b-hf	7B	apache-2.0	Standard (24GB)	llava-v1.6-mistral-7b-hf
	7B	apache-2.0		

Model	Parameters	License	GPU Requirement	Link
Qwen2-VL-7B-Instruct			Standard (24GB)	Qwen2-VL-7B-Instruct
Phi-3-vision-128k-instruct	4.2B	mit	Entry (16GB)	Phi-3-vision-128k-instruct

⭐ *= Recommended Clarifai model, ready to use. Check all models at clarifai.com/explore*

GPU Requirements Summary

Based on the recommended models:

- **Standard (24GB)**: mm-poly-8b, llava-1.5-7b-hf, llava-v1.6-mistral-7b-hf, Qwen2-VL-7B-Instruct
- **Standard (24GB) or Performance (48GB)**: Llama-3.2-11B-Vision-Instruct
- **Entry (16GB) or CPU**: all-MiniLM-L6-v2
- **Entry (16GB)**: bge-large-en-v1.5, Phi-3-vision-128k-instruct

Clarifai Platform Capabilities

Capability	Use For	Documentation
Model Library	Pre-built VLMs, LLMs, CV models	clarifai.com/explore
Workflows	Multi-model pipelines	docs.clarifai.com/create/workflows
AI Agents	Autonomous multi-step tasks	docs.clarifai.com/compute/agents
	RAG, semantic search	docs.clarifai.com/create/search

Capability	Use For	Documentation
Vector Search		
Pipelines	Async MLOps workflows	docs.clarifai.com/compute/pipelines

Recommended Approach

1. Start with Model Evaluation

Before reserving GPUs, evaluate Vision-Language Models (VLMs) against your data: - Start with Clarifai's mm-poly-8b for vision tasks - Test 2-3 alternative models for comparison - Measure accuracy on your specific use cases

Model Size → GPU Selection

Choose your GPU tier based on the model size you need:

Model Size	VRAM (FP16)	VRAM (INT8)	Recommended GPU	Examples
1-3B	6-8 GB	3-4 GB	T4 16GB	Small VLMs, embedders
7-8B	14-16 GB	7-8 GB	T4/L4 16-24GB	Llama 3.1 8B, Mistral 7B
13-14B	26-28 GB	13-14 GB	L4/L40S 24-48GB	Llama 2 13B
30-34B	60-70 GB	30-35 GB	L40S/A100 48-80GB	CodeLlama 34B
70B	140 GB	70 GB	A100 80GB or Multi-GPU	Llama 3.1 70B
100B+	200+ GB	100+ GB		Large models

Model Size	VRAM (FP16)	VRAM (INT8)	Recommended GPU	Examples
			Multi-GPU (2-8x H100)	

Tip: Use INT8 quantization to reduce VRAM ~50% with minimal quality loss.

2. Integrate via Clarifai API

Use Clarifai's managed API - no custom infrastructure needed:

```
from clarifai.client import Model

model = Model(url="https://clarifai.com/user/app/models/model-id")
response = model.predict(inputs=[your_data])
```

Documentation: https://docs.clarifai.com/compute/inference/

SDK Options: - Python: `pip install clarifai` (recommended) - Node.js: `npm install clarifai` - OpenAI-compatible: `https://api.clarifai.com/v2/ext/openai/v1` - LiteLLM, Vercel AI SDK also supported

3. Reserve GPU Capacity

Based on your workload, reserve appropriate capacity:

Phase	GPUs	Tier	Monthly Cost
Pilot (Months 1-3)	1-2	Standard (24GB)	~$1,469
Production (Months 4-6)	2-3	Standard/Performance	~$3,396
Scale (Months 7-12)	3-4+	Performance (48GB)	~$6,793

Estimated Year 1 Investment: ~$55,351

GPU Tier Reference

Tier	VRAM	Annual Cost	Best For
Entry	16GB	~$5,000-6,000	Development, testing
Standard	24GB	~$8,000-13,000	Production inference
Performance	48GB	~$18,000-25,000	LLM inference, high throughput
Enterprise	80GB+	~$21,000-65,000	Large models, enterprise scale

Compute Orchestration Benefits

(Full details)

- **Auto-scaling**: Scale from zero to infinity based on demand
- **GPU Fractioning**: Run multiple models per GPU
- **Model Packing**: Up to 3.7x compute efficiency
- **Cost Savings**: 60-90% savings vs. raw cloud GPUs

What Clarifai Handles

- ✅ GPU provisioning and management
- ✅ Infrastructure scaling (auto-scale to zero)
- ✅ Model hosting and serving
- ✅ API availability and uptime
- ✅ Security and compliance
- ✅ Updates and maintenance

What You Do

- ✅ Select models from Clarifai's library
- ✅ Integrate SDK/API into your application
- ✅ Reserve GPU capacity based on needs

- ✅ Monitor usage and adjust as needed

Next Steps

1. **Model Evaluation** - Test VLMs on your data (start with mm-poly-8b)
2. **Integration** - Use Clarifai SDK (Python, Node.js, or REST)
3. **GPU Reservation** - Reserve capacity based on evaluation results
4. **Go Live** - Deploy and scale as needed

Resources: | Resource | Link | |----------|------| | API Docs | https://docs.clarifai.com/ | | Model Library | https://clarifai.com/explore | | Inference | https://docs.clarifai.com/compute/inference/ | | Workflows | https://docs.clarifai.com/create/workflows/ | | Compute Orchestration | https://docs.clarifai.com/compute/overview |

Technical Architecture Recommendation

Hirtz & Carter – Automated Image-QA Solution

Industry: Retail (e-commerce)
Date: 15 Jan 2026

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1. Architecture Overview

Goal	**Detect image-quality problems (background, colour, retouching, compliance) automatically, return a structured JSON verdict with a human-readable explanation, and feed the result back to the existing DAM / CMS.**
Core Principle	**Vision-Language Model (VLM)** for high-level reasoning **+** lightweight detection models only when visual localisation is required.

Goal	**Detect image-quality problems (background, colour, retouching, compliance) automatically, return a structured JSON verdict with a human-readable explanation, and feed the result back to the existing DAM / CMS.**
Compute Model	Clarifai **Compute Orchestration** – managed GPU pool, auto-scaling, reservation option for predictable spend.
Key Benefits	• Explainable decisions (text only – VLM limitation) • No custom infra – everything runs on Clarifai SaaS • Scalable from 10 k to > 1 M images / month • Cost-optimised by reserving a modest GPU fleet.

1.1 High-Level Diagram

```
graph TD
    A[Source DAM / S3 / CMS] --> B[Ingestion Service (Clarifai SDK / webh
    B --> C[Pre-process (Resize, Validate, Metadata enrichment)]
    C --> D[Stage-1: VLM (mm-poly-8b)]
    D --> E{VLM flags "needs visual localisation"?}
    E -- Yes --> F[Stage-2: Detection Models]
    E -- No  --> G[Aggregate JSON Verdict]
    F --> G
    G --> H[Post-process (format, confidence thresholds)]
    H --> I[Webhook / REST API to DAM / CMS]
    H --> J[Dashboard / Reporting (Grafana/Clarifai Insights)]
```

All arrows represent asynchronous Clarifai workflow steps; each step can run in parallel on a shared GPU pool.

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2. Data Flow

Stage	Description	Formats / Tools
Ingestion	Pull new images from the customer's DAM (REST API), S3 bucket, or direct upload via Clarifai SDK.	JPEG/PNG/TIFF/PSD (max 5 MB after pre-resize).
Pre-processing	• Convert to RGB • Resize to ≤1024 px (preserves VLM quality) • Validate file integrity, colour profile (RGB).	Clarifai **pre-process operator** (built-in).
Stage-1 VLM	Single-pass multimodal inference that reads the image *and* the textual QA checklist (prompt). Returns: - `overall_status` (PASS/FAIL) - `defect_list` (array of issue codes) - `explanation` (natural-language rationale).	Model: **mm-poly-8b** (frontier VLM).
Conditional Stage-2	If VLM returns a "needs localisation" flag (e.g., *shadow detected*), invoke: - **Background-Detector** (object detector) – returns bounding box & confidence for background-colour deviation, shadows, stray objects. - **Colour-Accuracy-Classifier** – verifies colour match against reference swatches.	Models: `clarifai-retail-bg-detector` (YOLO-style) + `clarifai-color-accuracy-cls`.
Aggregation		

Stage	Description	Formats / Tools
	Merge VLM text verdict with any detection outputs into a single JSON payload.	Custom **workflow operator** (simple Python script).
Delivery	POST JSON to the customer's webhook endpoint; also write to a **Clarifai dataset** for audit and future retraining.	REST webhook, optional Kafka / SQS for downstream pipelines.
Reporting	Real-time dashboards (through Clarifai Insights) show pass-rate, defect categories, latency, and cost.	Grafana-compatible metrics, exported to CSV/Excel on demand.

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3. Clarifai Components

Component	Why Chosen	Configuration
mm-poly-8b (Frontier Multimodal)	- Handles complex visual reasoning (e.g., "Is lighting natural?"). - Returns **textual explanations** (VLM output limitation). - No need for model fine-tuning for the first 90 % of checklist items.	- Prompt template includes the full QA checklist (≈ 200 tokens). - Batch size = 4–8 per GPU for throughput.
clarifai-retail-bg-detector (Object Detection)	- Provides bounding-box evidence for background-colour, shadows, stray objects	- Confidence threshold = 0.65. - Runs only when VLM flags

Component	Why Chosen	Configuration
	– the only visual localisation the VLM cannot output.	`needs_localisation = true`.
clarifai-color-accuracy-cls (Classification)	- Quick, low-latency check that colour matches brand swatches (important for colour-accuracy checklist).	- Uses embedded reference colour vectors stored in a Clarifai dataset.
Workflow (`image_qa_pipeline`)	Orchestrates the three steps, handles branching, retries, dead-letter routing.	- Sequential: Pre-process → VLM → Conditional detection → Aggregation → Webhook.
Dataset "image_qa_ground_truth"	Stores every image + VLM verdict + detection results for audit and future fine-tuning.	- Auto-append via workflow operator.
Agents (optional)	If future "suggest corrective action" is required, an **MCP agent** can be added to call image-editing tools.	Not in MVP.

Custom Training? – Not required for MVP. If after evaluation the VLM misses >5 % of critical defects, a **fine-tune** on the 5 k labelled images can be performed (Clarifai Custom Model service) – budgeted for Phase 2.

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4. Integration Design

Aspect	Details
Incoming API	`POST /v2/users/{user_id}/apps/{app_id}/inputs` (Clarifai SDK) – bulk upload of image URLs or base64.
Outgoing API	Customer-provided webhook: `POST https://<customer>/image-qa-callback` with JSON payload: `{ "image_id": "...", "status": "PASS", "defects": [...], "explanation": "...", "detections": [{ "type":"shadow", "bbox": [x1,y1,x2,y2], "confidence":0.78 }] }`
Authentication	Clarifai **Personal Access Token (PAT)** for inbound calls. Customer webhook uses **HMAC-SHA256 signature** (shared secret) – verified by the customer's server.
Error Handling	- **Retry policy**: 3× exponential back-off for webhook failures. - **Dead-letter queue** (Clarifai "failed-events" dataset) for manual review. - **Workflow fallback**: if VLM returns an error, automatically route to a *fallback* simple classifier (`zero-shot-image-classifier`) to give a minimal PASS/FAIL.
SDK Samples	Python example for bulk upload & webhook registration (provided in the Implementation Guide).
Versioning	Model version stored in workflow metadata; change logs kept in a `model_registry` dataset.

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5. Infrastructure

5.1 Compute Requirements

Metric	Assumption	Result
Peak image rate	300 imgs / min (= 5 imgs / s)	
VLM latency	1.2 s per image on a **Standard-tier** GPU (NVIDIA A10, FP16)	
GPU-seconds per image	1.2 GPU-s	
Total GPU-seconds per second	5 imgs × 1.2 = 6 GPU-s	
GPUs needed (continuous)	6 GPU-seconds / 1 GPU-second ≈ 6 GPUs (Standard tier)	
Detection models	Run only on ~30 % of images, each ≈ 0.4 s, negligible extra capacity.	
Utilisation pattern	• Steady-state 5 imgs/s → 6 GPUs fully utilised. • Off-peak (night) < 1 img/s → utilisation drops to < 15 %.	
GPU type	**Standard tier** (24 GB VRAM, A10) – sufficient for **mm-poly-8b** (requires ~14 GB FP16 + 20 % KV cache).	
Reservation vs On-Demand	• On-Demand price (approx.) = $0.008 / GPU-sec → 6 GPUs × 86400 s × 30 days ≈ $124,416 / month. • 12-month **reserved capacity** (30 % discount) ≈ $87,091 / month → **$1.05 M / year**. • If peak usage only 30 % of the month, a **mixed**	

Metric	Assumption	Result
	model (reserve 3 GPUs, burst up to 6 on-demand) reduces annual spend to **≈ $720 k**.	
ROI Calculation	- **Manual QA cost**: 2 FTE × $80 k / yr ≈ $160 k (250 imgs/FTE-day). - **Re-work cost saved**: 30 % reduction on $22 k / mo ≈ $79 k / yr. - **Total annual benefit** ≈ $239 k. - **Net cost** (mixed reservation) ≈ $720 k → **Payback ≈ 3.0 yr**. - If throughput can be reduced to 2 imgs/s (e.g., batch-only at night) the required GPUs drop to 2-3, cutting cost to **≈ $350 k / yr**, **payback < 2 yr**.	
Recommendation	Start with a **mixed reservation**: **3 Standard-tier GPUs reserved** (covers average load) + **burst pool of up to 6 GPUs on-demand** for peak spikes. This gives a **~40 % cost saving** vs pure on-demand while guaranteeing SLA.	

5.2 Deployment Topology

```
+-------------------+      +--------------------+      +-----------------
|   Customer DAM / | ---> |   Clarifai Ingest   | ---> |  Clarifai GPU Po
|   S3 / CMS       |      |   (SDK / webhook)   |      | (Standard-tier)
+-------------------+      +--------------------+      +----------+------
                                                          |
                                                          v
                                                  +-------------------+
                                                  |  Workflow Engine  |
                                                  +-------------------+
                                                          |
                  +-------------------+--------------------+--------------
                  |                   |                    |
```

```
                v                    v                     v
          VLM (mm-poly-8b)   Detection (bg-det)   Detection (colour)   Ag
                |                    |                     |
                +--------------------+---------------------+--------------
                                                          |
                                                          v
                                                +-------------------+
                                                |   Webhook / API   |
                                                +-------------------+
```

All services are fully managed by Clarifai (SaaS). No customer-owned servers needed.

5.3 High-Availability

- **GPU pool** – Clarifai automatically spreads workloads across the reserved GPU nodes; if a node fails, the scheduler re-routes jobs to the burst pool.
- **Workflow retries** – built-in exponential back-off (max 5 attempts).
- **Data durability** – Input images stored in Clarifai's immutable "input" store; results persisted in a dedicated dataset with 7-day versioning.

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6. Security

Area	Controls
Data in transit	TLS 1.2+ for all API calls (both inbound to Clarifai and outbound webhook).
Data at rest	Images stored encrypted (AES-256) in Clarifai's object store; customer can enable **customer-managed keys (CMK)** if required.
Authentication	PAT for Clarifai SDK; HMAC-SHA256 signature for outbound webhook.

Area	Controls
Authorization	Role-Based Access Control (RBAC) within Clarifai app – only QA engineers can view raw images; analysts can view only JSON verdicts.
Compliance	GDPR-compliant data handling; can be deployed in **EU-region** VPC if needed.
Audit	All inference calls logged with user, timestamp, model version; exportable audit logs for compliance teams.

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7. Monitoring & Observability

Metric	Source	Target / Alert
Requests / sec	Workflow engine	>5 req/s → scale-up alert.
Average latency (VLM)	GPU utilization logs	>2 s → investigate GPU saturation.
GPU utilization %	Clarifai Compute Dashboard	>85 % → auto-scale burst pool.
Error rate (inference failures)	Workflow dead-letter queue	>0.5 % → raise ticket.
Pass-rate (overall_status)	Dataset "image_qa_ground_truth"	Sudden drop >10 % → QA rule drift alert.
Cost per 1 k images	Billing API	>$12 → cost-optimisation review.
Webhook delivery success	Outbound webhook logs	<99 % success → retry escalation.

All metrics are visualised in a Grafana dashboard (Clarifai Insights integration). Alerts are routed to Slack, PagerDuty, or email per the customer's preference.

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8. Implementation Phases

Phase	Duration	Key Deliverables	Dependencies
0 – Kick-off & Requirements	1 wk	Signed requirements doc, access to DAM, webhook URL, reference QA checklist.	Customer provides PAT, webhook secret.
1 – Data Prep & Ground-Truth Set	2 wks	5 k labelled images (PASS/FAIL + defect tags) stored in `image_qa_ground_truth`.	Access to historical QA logs.
2 – Model Evaluation	3 wks	Benchmark report for `mm-poly-8b` vs 2 alternatives (e.g., `LLaVA-1.5-8b`, `open-clip-v2`). Decision matrix.	Compute budget for evaluation (on-demand).
3 – Workflow Build (MVP)	2 wks	Working Clarifai workflow (`image_qa_pipeline`) in **staging**; webhook test harness.	Model versions selected.
4 – Integration & UI	2 wks	SDK wrapper for bulk upload, webhook consumer, simple React dashboard (optional).	Customer's CMS/DAM API.
5 – Performance	1 wk	GPU count finalised, reservation plan signed, auto-scale thresholds set.	Billing approval.

Phase	Duration	Key Deliverables	Dependencies
Tuning & Scaling			
6 – UAT & Sign-off	1 wk	End-to-end test with 10 k live images, accuracy ≥90 % on critical defects, latency ≤2 s.	QA team validation.
7 – Production Roll-out	1 wk	Reserved GPUs provisioned, workflow switched to prod, monitoring enabled.	Change-management approval.
8 – Phase-2 Enhancements (optional)	4 wks	Fine-tune VLM on H&C data, add “suggest corrective action” agent, integrate with ticketing system.	Positive MVP metrics.

Risk Mitigation

Risk	Mitigation
VLM latency exceeds SLA	Batch inference, reserve extra GPU headroom, fallback to lightweight classifier for non-critical images.
Insufficient labelled data	Use synthetic augmentation (lighting, background swaps) + active-learning loop (human-in-the-loop for low-confidence cases).
Data-privacy restrictions	Deploy Clarifai **private VPC** in the EU region; enable CMK encryption.
Model drift	Monthly re-evaluation against ground-truth set; automated retraining pipeline (Phase-2).
Budget overruns	

Risk	Mitigation
	Start with mixed reservation (3 GPUs reserved) and monitor; adjust reservation up/down quarterly.

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9. Summary & Recommendations

What	Why	Action